

Performance Testing

Project Title

AI Learning Assistant

Domain

Artificial Intelligence | Education Technology

1. Introduction

Performance Testing evaluates how efficiently the AI Learning Assistant performs under normal operating conditions. It ensures that the application responds quickly, remains stable, and provides a smooth user experience.

2. Objectives

The objectives of performance testing are:

- Verify response time.
 - Ensure system stability.
 - Check AI response generation.
 - Validate dashboard performance.
 - Measure overall usability.
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3. Test Environment

Component	Details
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Operating System	Windows 11
Programming Language	Python 3.10+
IDE	Visual Studio Code
Framework	Streamlit
AI Model	Google Gemini AI
Data Storage	CSV File
Browser	Google Chrome

4. Test Cases

Test Case	Expected Result	Status
Launch Application	Application opens successfully	✔ Pass
Enter User Input	Input accepted without errors	✔ Pass
Emotion Detection	Correct emotion identified	✔ Pass
AI Response Generation	Personalized response generated	✔ Pass

CSV Logging	Interaction stored successfully	✓ Pass
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Dashboard Loading	Analytics displayed correctly	✓ Pass
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5. Performance Metrics

Parameter	Result
Application Startup	Fast
Emotion Detection	Accurate
AI Response Time	Within a few seconds
Dashboard Performance	Smooth
Memory Usage	Efficient

6. Observations

- The application responded quickly.
 - Emotion detection worked correctly for supported emotions.
 - Gemini AI generated meaningful responses.
 - Dashboard displayed analytics without errors.
 - CSV logging functioned properly.
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7. Conclusion

The AI Learning Assistant successfully passed all major performance tests. The system demonstrated good responsiveness, stability, and usability, making it suitable for educational applications.